Content Determination of Potassium Dihydrogen Phosphate

Method for the alkalimetric content determination of Potassium Dihydrogen Phosphate

Potassium dihydrogen phosphate is a saline laxative, which is used to treat hyperpotassaemia from poor or malnourishment, or cases of high loss of potassium from vomiting or diarrhoea.

Sample	Potassium dihydrogen phosphate
Substance	Potassium dihydrogen phosphate M = 136.1 g/mol
Chemicals	Deionised water R
Titrant	c(NaOH) = 0.5 mol/l
Standard	Potassium hydrogen phthalate (M521)
Instruments	METTLER TOLEDO DL70ES/DL77 METTLER TOLEDO DL58/55/53/50Graphix
Accessories	Titration beaker ME-101974 SP250 Peristaltic pump Rondo60 Sample changer
Indication	DG111-SC
Chemistry	Deprotonation of the dihydrogen phosphate anions to monohydrogen phosphate $H_2PO_4^- + OH^- \rightarrow HPO_4^{2-} + H_2O$
Calculation	R(%) = (Q • M/(10 •m)) + H20 Q = Titrant consumption in mmol M = Molar mass K-dihydrogen phosphate m = Sample size in g H20 = % Loss on drying of KH_2PO_4

Structure	
Literature	Ph. Eur. 1997; Monograph "Potassium dihydrogen phosphate"

Sample preparation

Weigh a sample of 0.24 to 0.28 g potassium dihydrogen phosphate, accurately to 0.1 mg, into a 100 mL titration beaker and dissolve in 40 mL deionised water.

Remarks

The previously determined loss on drying content is saved as the auxilliary value H20.

Waste disposal

Halogen-free organic solvents waste.

Author

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Results

METTLER DL77 Titrator V3.1 Tante Elfriede
Dr. Kade Pharm. Fabrik GmbH Quality Control Laboratory

Method: G056 Cont. KH2PO4 Date: 14-12-1999 13:40
User: **Pludra** Measured: 14-12-1999 13:43

RESULTS

No.	ID1	Sample size and results
1/1	992222	0.2564 g R1 = 100.403 % KH2PO4
1/2	992222	0.2453 g R1 = 100.270 % KH2PO4
1/3	992222	0.2443 g R1 = 100.278 % KH2PO4
1/4	992222	0.2581 g R1 = 100.294 % KH2PO4
1/5	992222	0.2471 g R1 = 100.243 % KH2PO4
1/6	992222	0.2553 g R1 = 100.171 % KH2PO4

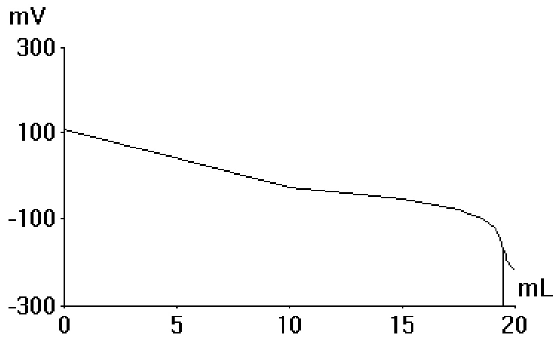
STATISTICS

Number of results R1	n = 6	
Mean value	$\bar{x}$ = 100.276	% KH2PO4
Standard deviation	s = 0.075597	% KH2PO4
Rel. standard deviation	srel = 0.075	%

Table of measured values

EQP	ET	Volume mL	Signal mV	1st Deriv. mV/mL	Time min:s
	ET1	0.0000	108.1		0:02
		9.9820	-25.7	-13.4	0:27
		14.9730	-54.5	-5.8	1:10
	ET2	17.4700	-78.7	-9.7	1:21
		17.9700	-86.8	-16.1	1:28
		18.3250	-94.3	-21.1	1:35
		18.6310	-102.5	-27.0	1:42
		18.8740	-111.4	-36.3	1:49
		19.0490	-120.2	-50.4	1:56
		19.1730	-128.5	-67.2	2:03
		19.2680	-137.0	-89.2	2:11
		19.3390	-145.5	-119.3	2:20
		19.3920	-153.2	-145.3	2:28
		19.4380	-160.2	-152.2	2:36
EQP1		19.4880	-168.8	-173.6	2:43
		19.5290	-175.8	-170.7	2:51
		19.5760	-183.2	-156.4	2:58
		19.6320	-190.9	-137.5	3:06
		19.6990	-198.0	-105.5	3:13
		19.8070	-206.9	-83.0	3:20
		19.9390	-215.2	-63.1	3:27
		20.0000	-218.4	-51.6	3:33

Titration curve



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Method:     G056     Cont. KH2PO4
             Version:  14-12-1999  13:40

Title
   Method ID  . . . . . . . . . . . . . G056
   Title  . . . . . . . . . . . . . . . Cont. KH2PO4
   Date/time  . . . . . . . . . . . . . 14-Dec-1999  13:29
Sample
   Number samples . . . . . . . . . . . 6
   Titration stand  . . . . . . . . . . ST20 1
   Entry type . . . . . . . . . . . . . Weight m
      Lower limit [g] . . . . . . . . . 0.24
      Upper limit [g] . . . . . . . . . 0.26
   ID1  . . . . . . . . . . . . . . . .
   Molar mass M . . . . . . . . . . . . 136.1
   Equivalent number z  . . . . . . . . 1
   Temperature sensor . . . . . . . . . Manual
Stir
   Speed [%]  . . . . . . . . . . . . . 50
   Time [s] . . . . . . . . . . . . . . 10
Titration
   Titrant  . . . . . . . . . . . . . . NaOH
   Concentration [mol/L]  . . . . . . . 0.1
   Sensor . . . . . . . . . . . . . . . DG111-SC
   Unit of meas.  . . . . . . . . . . . mV
   Titration mode . . . . . . . . . . . EQP
      Predispensing 1 . . . . . . . . . % nom. content
         Metered amount [%] . . . . . . 90.0
         Nominal content  . . . . . . . 100.0
         Conversion const.  . . . . . . M/(10*z)
      Titrant addition  . . . . . . . . DYN
         dE(set) [mV] . . . . . . . . . 8.0
         Limits dV  . . . . . . . . . . Absolute
            dV(min) [mL]  . . . . . . . 0.02
            dV(max) [mL]  . . . . . . . 0.5
      Measure mode  . . . . . . . . . . EQU
         dE [mV]  . . . . . . . . . . . 0.5
         dt [s] . . . . . . . . . . . . 1.0
         t(min) [s] . . . . . . . . . . 2.0
         t(max) [s] . . . . . . . . . . 20.0
      Threshold . . . . . . . . . . . . 50.0
      Maximum volume [mL] . . . . . . . 20.0
      Evaluation procedure  . . . . . . Standard
      Steepest jump only  . . . . . . . Yes
      Stop for reevaluation . . . . . . Yes
         Condition  . . . . . . . . . . neq=0
Calculation
   Result name  . . . . . . . . . . . . KH2PO4
   Formula  . . . . . . . . . . . . . . R=(Q*C/m)+H20
   Constant . . . . . . . . . . . . . . C=M/(10*z)
   Result unit  . . . . . . . . . . . . %
   Decimal places . . . . . . . . . . . 3
Rinse
   Auxiliary reagent  . . . . . . . . . Water
   Volume [mL]  . . . . . . . . . . . . 30.0
Record
   Output unit  . . . . . . . . . . . . Computer
   Raw results last sample  . . . . . . Yes
   Table of values  . . . . . . . . . . Yes
   E - V curve  . . . . . . . . . . . . Yes
Statistics
   Ri (i=index) . . . . . . . . . . . . R1
   Standard deviation s . . . . . . . . Yes
   Rel. standard deviation srel . . . . Yes
Record
   Output unit  . . . . . . . . . . . . Printer
   Short-form method  . . . . . . . . . Yes
   All results  . . . . . . . . . . . . Yes
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